

PROJECT DOCUMENTATION

PROJECT DESIGN PHASES

EduGenie – Google Gemini Powered Learning Assistant

1. Design Goal

Design a simple and student-friendly AI learning interface that combines several learning activities in one web application.

2. Architecture Design

The system is divided into Frontend Layer, Application Layer, AI Layer, Configuration Layer, and Template Layer.

3. Interface Design

The interface contains a project header, navigation links, input areas, buttons, cards, and a dynamic AI answer section. CSS provides the visual layout and responsive design.

4. Module Design

The documented interface includes Ask, Explain, Quiz, Learning Path, and Summarize sections.

5. Backend Design

FastAPI manages routes and request handling. The /ask route accepts form data and sends the question to Gemini.

6. Data and Configuration Design

The current basic question-answering prototype does not require a traditional database. The Gemini API key is stored in a .env file and should not be exposed publicly.

7. Future Design

A production version could add user accounts, learning history, quiz scores, saved summaries, progress tracking, multilingual support, voice interaction, mobile applications, and LMS integration.

Prepared for project documentation and academic demonstration.